

Current and Resistance

1 Introduction

1.1 Definitions

The electric current in a wire is defined as

$$I = \frac{dQ}{dt}$$

where dQ is the total amount of charge that passes through a cross-section of the wire in a differential amount of time, dt .

If q is the charge (in Coulombs) of each flowing charge, n is their number per volume (“number density”), v_d their average speed along the wire (called the “drift speed”), then

$I = n|q|v_dA$, where A is the cross-sectional area of the wire.

The average current density is defined as

$$J = \frac{I}{A}, \text{ which can be written as } J = n|q|v_d.$$

1.2 Ohm’s Law

If an electric field exists in a wire (by, for example, connecting its ends to a battery), the charges will accelerate until they collide with another particle and decelerate (collisions resist the flow). The net result will be current – a flow of charges with an average drift speed. Experimentally, it has been shown that in many materials, the electric field is proportional to the current density:

$$E = \rho J$$

where the value of the proportionality constant ρ , called resistivity, depends on the material. This is one version of Ohm’s law.

For a wire of length L and constant cross-sectional area A , Ohm’s law can also be written as $V = (\rho L/A)I$. If we define resistance as $R = \rho L/A$, then we have another relationship that is also referred to as Ohm’s law:

$$V = IR.$$

In this form, the interpretation is that the voltage between the ends of a wire is proportional to the current in the wire, with the proportionality constant of R .

2 Problems

2.1 Definitions and Ohm's Law

A 9-volt power source is connected to a wire of length 10 meters with a circular cross-section and radius of 0.01 meters. The wire has a resistivity of $10^{-8} \Omega \cdot \text{m}$. The number density of flowing charges is $10^{28} / \text{m}^3$. Assume Ohm's law applies and the charges that flow are electrons.

1. What is the resistance (with units) of the wire?
2. What is the current (with units) in the wire?
3. How much charge (in Coulombs) flows past a cross-section of the wire per second?
4. How many electrons flow past a cross-section of the wire per second? (The charge on an electron is $-1.6 \cdot 10^{-19} \text{ C}$.)
5. What is the current density (with units)?
6. What is the drift velocity of electrons in the wire? (The charge on an electron is $-1.6 \cdot 10^{-19} \text{ C}$.)

7. Based on the description of how charged particles flow in a wire, explain why the resistance of a cylindrical wire is proportional to its length and inversely proportional to the square of its radius.

2.2 Definitions and Ohm's Law

A power source is connected to a 20 meter long wire with a circular cross-section and radius of 0.01 meters. The wire has a resistivity of $10^{-7} \Omega \cdot \text{m}$. The number density of flowing charges is $10^{27}/\text{m}^3$. The current in the wire was measured and found to be 1 A. Assume Ohm's law applies and the charges that flow are electrons.

1. What is the resistance (with units) of the wire?
2. How much charge (with units) flows past a cross-section of the wire per second?
3. What is the current density (with units)?
4. What is the drift velocity of electrons in the wire? (The charge on an electron is $-1.6 \cdot 10^{-19} \text{ C}$.)

2.3 Current Through a Cylindrical Shell

If a cylindrical wire with an inner radius a and outer radius b carries a current I , what is J ?

2.4 $I = n|q|v_dA$ derivation

Derive the relationship $I = n|q|v_dA$. Provide a diagram.